

The process of Electrolysis

- What electrolysis is and the reactions
- Explain why a mixture is used and the anode needs replacing
- Write and balance half equations for the electrode reactions



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School

Key words

electrolysis

electrolytes

electrode

Ion Migration

Retrieval questions

1. Which one has only one type of atom in it?
a) element b) compound
2. Water is an example of a
a) mixture b) compound c) solution
3. Which of the following is a compound
a) Li b) Ca c) CuSO₄
4. What subatomic particle is involved in making a compound?
a) protons b) neutrons c) electrons
5. What ions do acids release in solution?
a) Hydrogen ions (H⁺) b) Hydroxide ions (OH⁻) c) Sodium ions (Na⁺) d) Chloride ions (Cl⁻)
6. What is the pH of acids?
a) above 7 b) below 7 c) 7

Retrieval questions

1. Which one has only one type of atom in it?
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2. Water is an example of a
a) mixture b) **compound** c) solution
3. Which of the following is a compound
a) Li b) Ca c) **CuSO₄**
4. What is the compound when sodium and chlorine react together
a) sodium chlorine b) **sodium chloride** c) sodium chlorate
5. What ions do acids release in solution?
a) **Hydrogen ions (H⁺)** b) Hydroxide ions (OH⁻) c) Sodium ions (Na⁺) d) Chloride ions (Cl⁻)
6. What is the pH of acids? above 7 b) **below 7** c) 7

Put these elements in the correct boxes

Copper carbonate

Lead

Gold

Silver

Iron

Iron oxide

Aluminium oxide

Sodium

Platinum

Native metals

Non-native elements

Ores

Native metals

Gold

Silver

Platinum

Non-native elements

Iron

Lead

Sodium

Ores

Copper carbonate

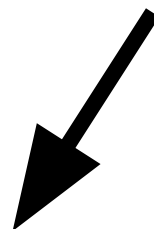
Iron oxide

Aluminium oxide

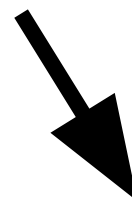
What electrolysis is

Electrolysis is the process of passing direct current (d.c.) through a solution or melted ionic compound to move the ions apart and so break the compound down and discharge some of the elements at the electrodes.

Electrolysis



Electricity



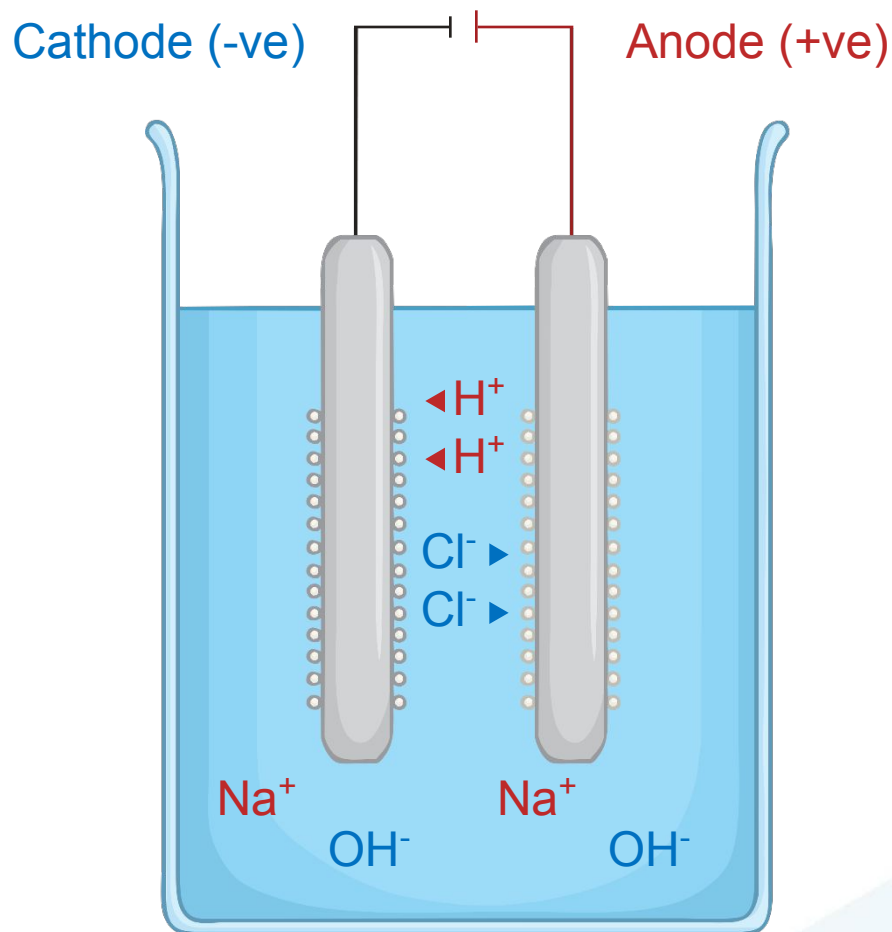
To break down

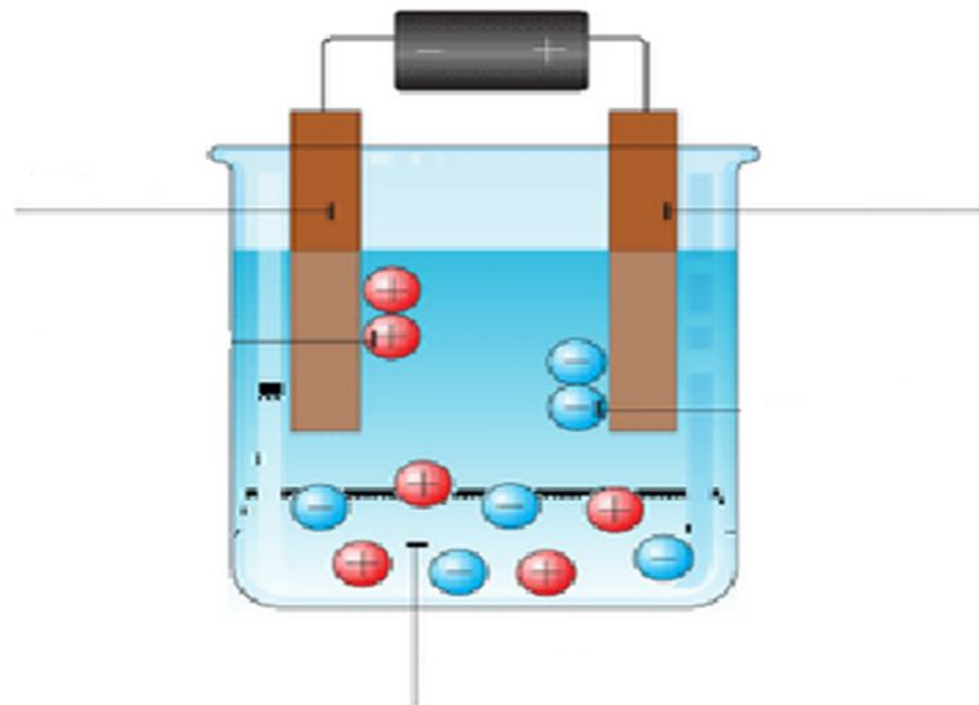
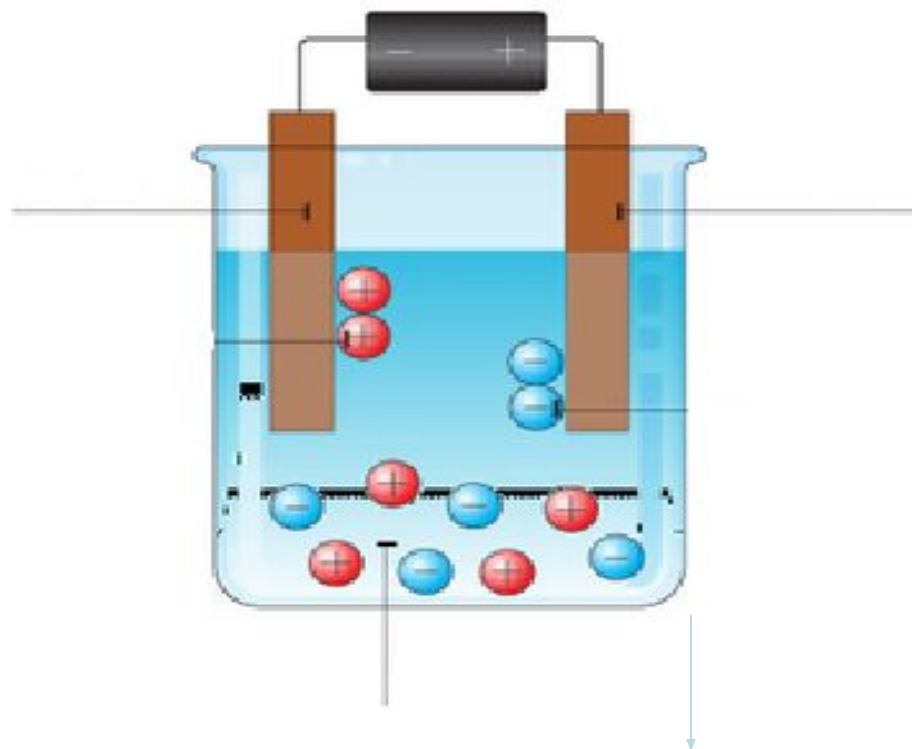
To use electricity to break down compounds

Predicting the Products of Electrolysis – Negative Electrode

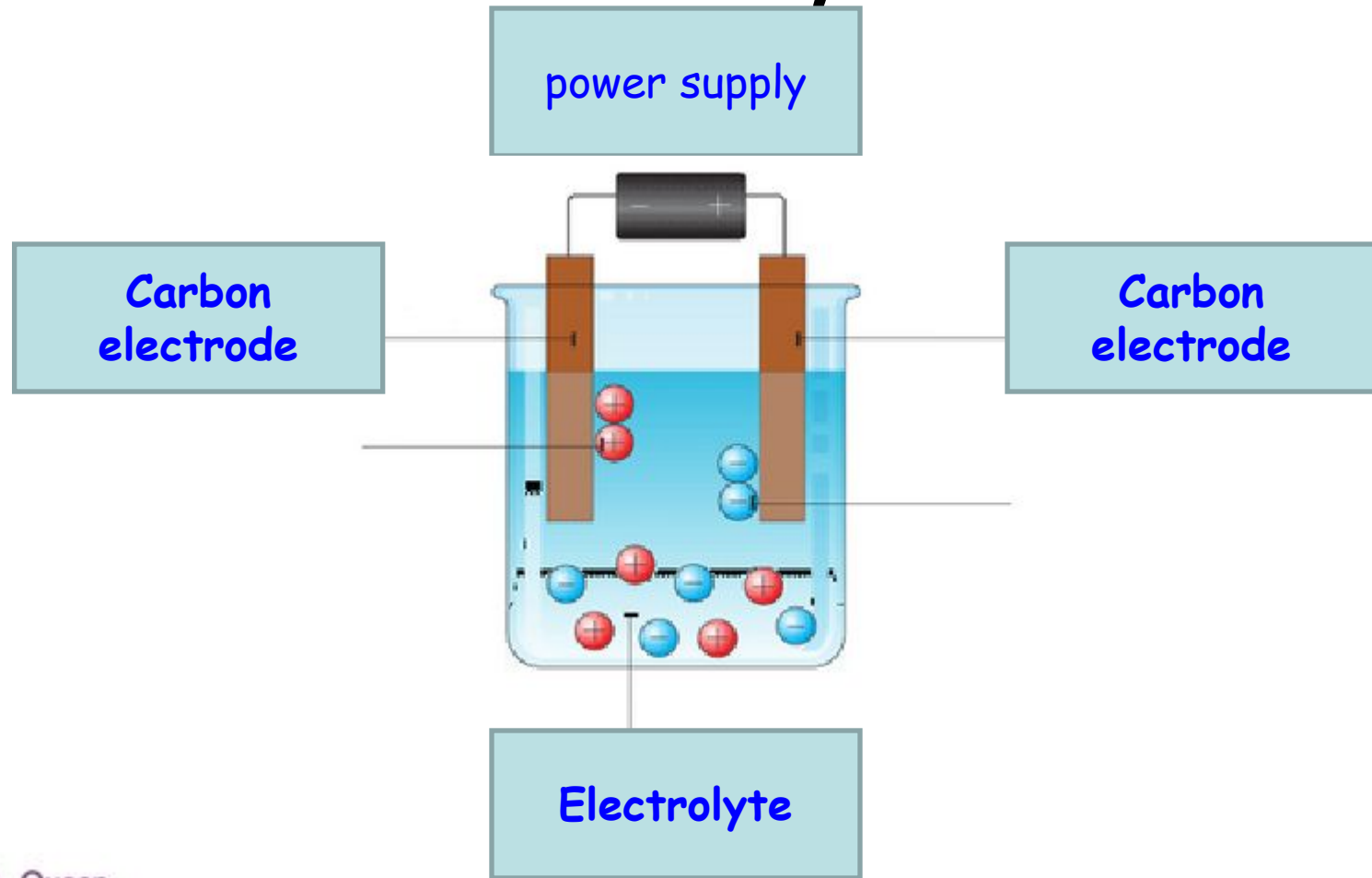
When an ionic compound undergoes electrolysis the positive and negative ions are attracted to the oppositely charged electrode where they may be oxidised (lose electrons) or reduced (gain electrons) to become the element again.

Predicting the Products of Electrolysis – Negative Electrode

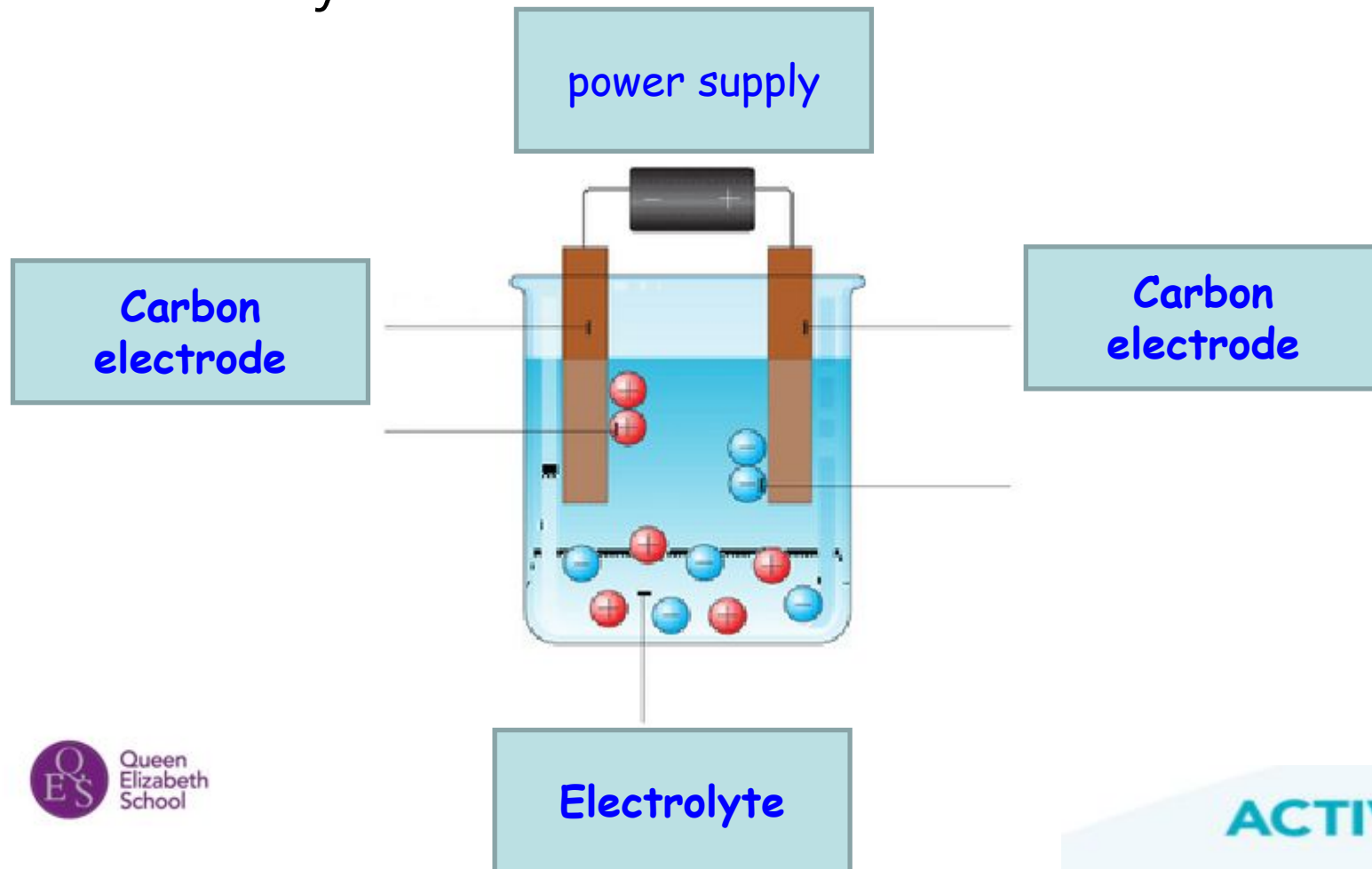




Equipment used to carry out Electrolysis



The solution or molten compound is called the electrolyte. The ions need to be free to move about within the solution or liquid to be an electrolyte. An electrolyte is a liquid that conducts electricity and is decomposed during the electrolysis.

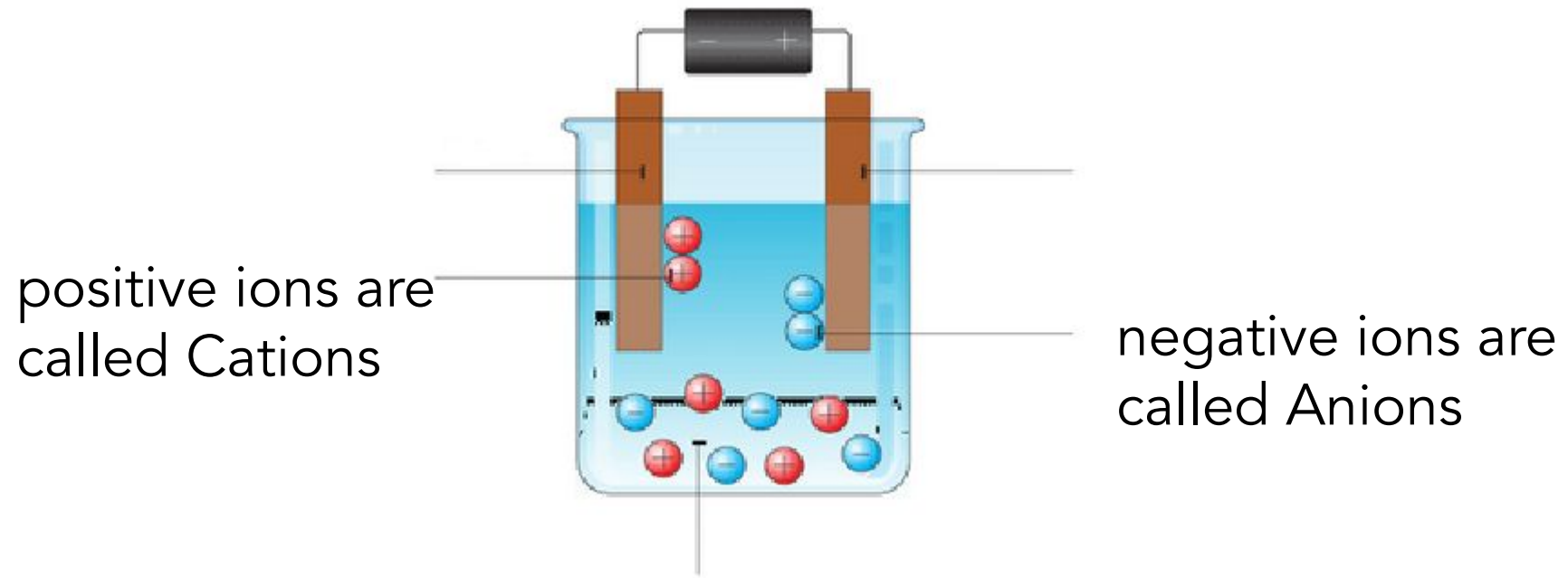


Activity to stick in books

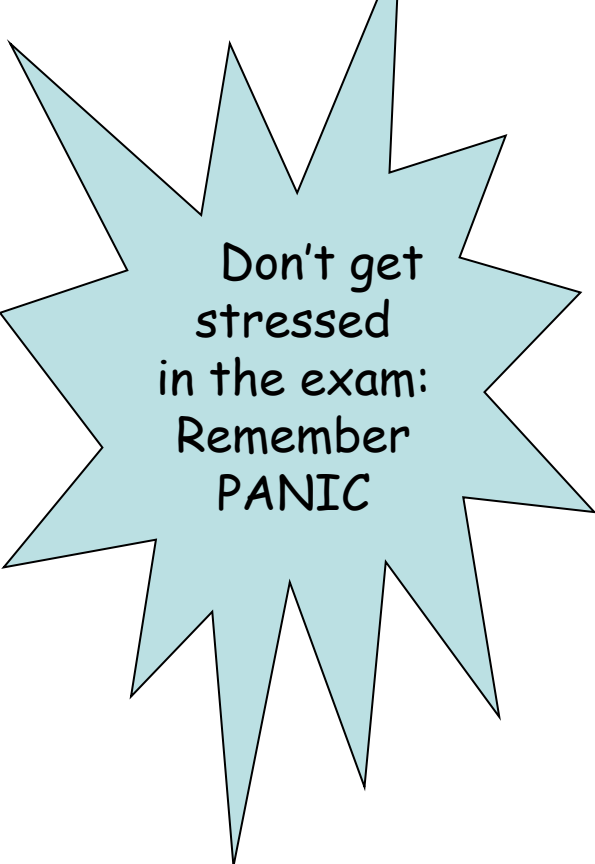
Electrolysis is a process that splits a _____ back to its original elements. It is based on an _____ circuit that includes a liquid which is called an _____ there are also two _____

In electrolysis, the electrodes are placed into the electrolyte and ions move from one electrode to the other _____ this allows the conduction of electricity through the circuit.

Electrolytes contain free ions — they're usually the molten or dissolved ionic substance. It would not be possible to conduct electricity without either melting or dissolving. In either case it's the free ions which conduct the electricity and allows the whole thing to work.



ACTIVATION



Don't get
stressed
in the exam:
Remember
PANIC

- Positive
- Anode
- Negative
- Is
- Cathode

On your whiteboards choose the correct one for each statement

1.
Cathode is
Negative/Positive?

2.
Anode is
Positive/negative?

3.
Positive/negative ions
attracted?

4. Negative/positive
ions attracted?

Recording Results

Solution	Negative electrode (cathode) observations and element present	Positive electrode (anode) observations and element present
Copper chloride		
Copper sulphate		
Sodium chloride		
Sodium sulphate		

complete the worksheet

Key questions

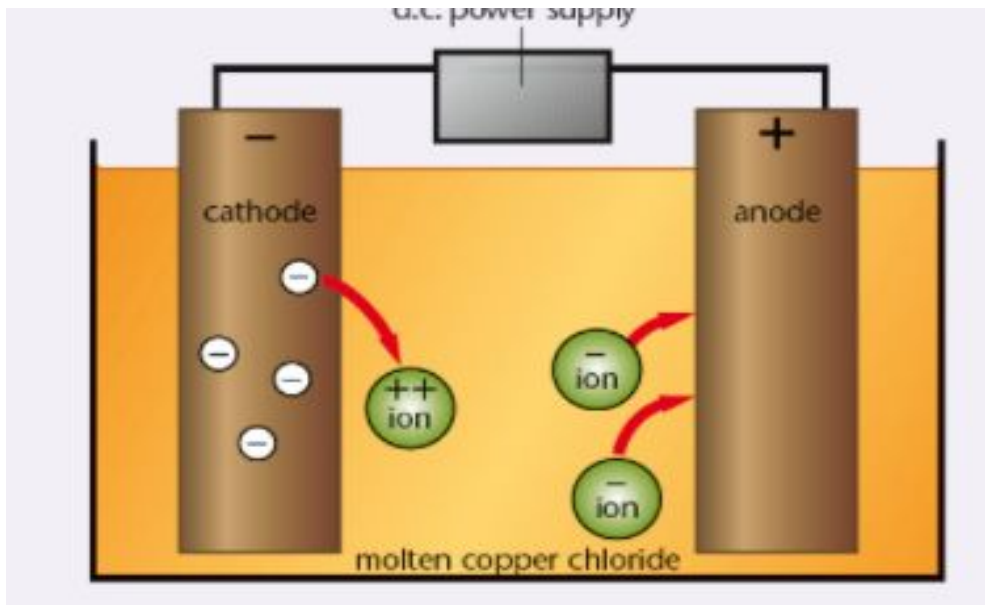
1. What happens to an ionic substance when it is melted or dissolved in water?
2. What is the name for the positive electrode?
3. What is the name for the negative anode?
4. Do positive ions move to the anode or the cathode?
5. Do negative ions move to the anode or the cathode?
6. At which electrode would $\text{Zn}^{2+}(\text{aq})$ turn into $\text{Zn}(\text{s})$?
7. At which electrode would $\text{Cl}^{-}(\text{aq})$ turn into $\text{Cl}_2(\text{g})$?

Key questions answers

1. The ions become free to move around
2. The anode
3. The cathode
4. Cathode
5. Anode
6. Cathode (needs to gain electrons)
7. Anode (needs to lose electrons)

Ions discharging

- All the **ions migrate** towards the 2 electrodes.
- All the positive ions are attracted to the negatively charged electrode.
- All the negative ions are attracted to the positively charged electrode.
- When the negative ions reach the anode they lose electrons to the electrode.



When the positive ions reach the cathode they gain electrons at the electrode

Write half equations

Li^+ Gains an electron $\rightarrow \text{Li}$

Mg^{2+} Gains two electrons $\rightarrow \text{Mg}$

Cl^- Lose one electrons $\rightarrow \text{Cl}$

O^{2-} Lose two electrons $\rightarrow \text{O}$

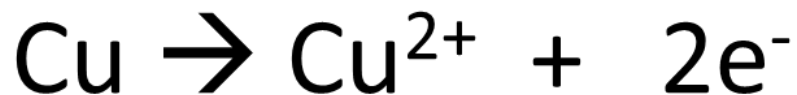
Write half equations

- We can write the loss or gain of electrons as an equation like this:

Sodium atoms (Na) lose 1 electron to become a sodium ion (Na⁺)



Copper atoms lose 2 electrons to become copper ions



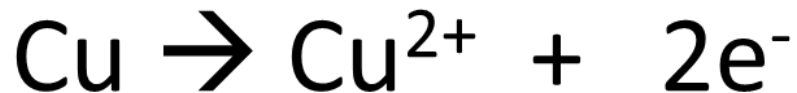
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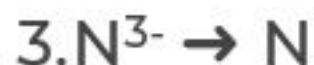


Copper atoms lose 2 electrons to become copper ions



Independent task

Describe what has happened in terms of electrons for the changes below to occur. State if electrons have been lost or gained and how many have been lost or gained.



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